

AeroSense Sensor Enclosure

Ceiling-Mounted Node — Design Options & Dimensions

Prepared: June 20, 2026

Contents: ESP8266 NodeMCU + PMS5003 + SGP40 (no Raspberry Pi)

Scope: one device, ceiling of a 9 ft (2.74 m) room

1. Overview & mounting context

This document specifies the enclosure for a single AeroSense sensor node mounted on the ceiling (roof) of a room about **9 ft (2.74 m)** tall. The node houses the **ESP8266 NodeMCU** (the Wi-Fi MCU that POSTs readings to the dashboard), the **PMS5003** particulate sensor, and the **SGP40** VOC sensor. The Raspberry Pi is intentionally **not** included: the ESP8266 has its own Wi-Fi and uploads directly, so no on-site gateway is needed. This keeps the box small.

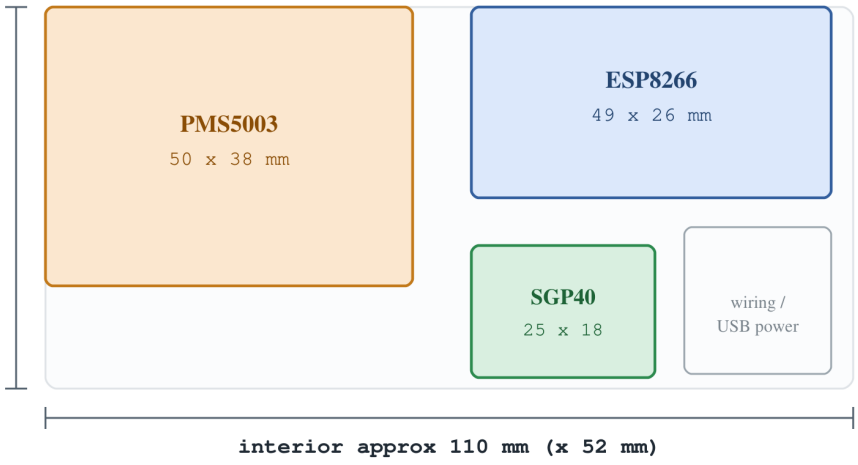
- **One device first.** Detection radius / coverage is deliberately out of scope for now; it only affects how many units and their spacing, not the box size.
- **Faces down into the room.** Vapor rises, so a ceiling position catches it well.
- **Air must reach the sensors.** The PMS5003 fan needs clear intake/exhaust and the SGP40 must sample room air, so vents sit on the face and/or sides - never flat against the ceiling.

2. What is inside & the minimum size

The three modules pack into roughly a **110 x 52 mm** interior footprint (about 21 mm tall for the PMS5003). That internal envelope, plus walls, standoffs, vents and wiring, is what drives the enclosure sizes in Section 3.

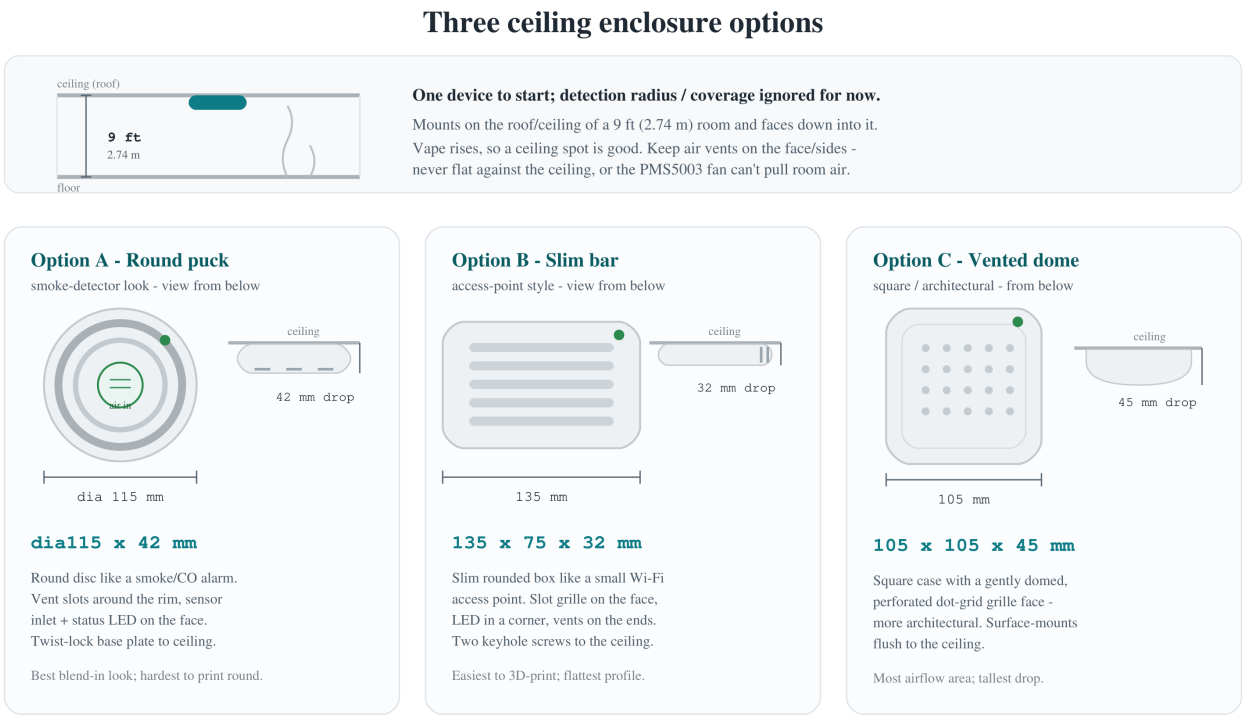
Component	Footprint	Height	Needs
ESP8266 NodeMCU	49 x 26 mm	~13 mm	Wi-Fi MCU; posts to the dashboard
PMS5003 (PM2.5)	50 x 38 mm	21 mm	clear air vents (fan intake/exhaust)
SGP40 (VOC)	25 x 18 mm	~6 mm	exposure to room air

Internal layout (top-down) - sets the minimum size



Internal top-down layout (no Raspberry Pi). Typical module sizes.

3. The three enclosure options



Three ceiling-mount appearance options, drawn to scale, each holding the same node.

Option	Look	Length x Width	Drop	Best for
A - Round puck	smoke-detector	115 x 115 mm (dia 115)	42 mm	blends into the ceiling
B - Slim bar	access-point	135 x 75 mm	32 mm	easiest to 3D-print; flattest
C - Vented dome	square / architectural	105 x 105 mm	45 mm	most vent area

All three comfortably hold the ESP8266 + PMS5003 + SGP40 with room for airflow and wiring. Sizes are typical and include ~5 mm walls plus margins; verify against your exact breakout boards (NodeMCU and SGP40 modules vary in size).

4. Mounting, airflow & power

Orientation & airflow

- The vented face points down into the room; keep the PMS5003 intake unobstructed.
- Do not seat the vents against the ceiling surface - leave a side/face air path.
- If the I2C/VOC reading is flaky, add 4.7 kohm pull-ups (SDA->3.3V, SCL->3.3V) on the board, not the case.

Power

- The node runs on **5 V USB**. On a ceiling, plan a cable run to an outlet, a USB adapter tucked above the tile, or a PoE-to-USB splitter.
- Add a small cable gland or notch in the case for the USB lead.

Mounting

- Option A: twist-lock base plate (screwed to a ceiling box/anchors), case clicks on.

- Option B: two keyhole slots - screw two anchors, hang and slide.
- Option C: surface-mount flange with two/four screws, flush to the ceiling.

5. Recommendation & next steps

For a capstone build, Option B (slim bar) is the most practical - it is the easiest to 3D-print, has the flattest profile, and gives the PMS5003 a clean slot-grille air path. **Option A (round puck)** looks the most natural on a ceiling (reads as a smoke alarm) if you can print/source a round shell. **Option C** offers the most vent area if airflow is a concern.

- Pick A, B, or C and a full dimensioned build drawing can follow (internal mounts, vent cut-outs, screw holes, cable gland).
- Bring back detection radius / coverage later to decide how many units and their spacing - it will not change this box size.